Disease Atlas and Ciaran Space Ontology Comparison

A practical comparison of the live Geo ontology used in Disease Atlas and the ontology currently used in Ciaran's metabolic-model space, with recommendations for reuse, provenance, source identifiers, and cross-space interoperability.

Prepared from live Geo queries on April 20, 2026.

Spaces reviewed: Disease Atlas https://www.geobrowser.io/space/141d3ace705feabc04d50c78bbf7226e and Ciaran's space https://www.geobrowser.io/space/aec359295ff90d00aea6696f0ddb1bbc.

Executive summary

Ciaran's space is a metabolic-network style graph, probably imported from BiGG / an E. coli K-12 MG1655 genome-scale model. Disease Atlas is a disease-mechanism and evidence graph.

The biggest ontology issue in Ciaran's space is not the metabolism-specific terms. Those are mostly reasonable. The biggest gaps are provenance and reuse:

- Gene is duplicated locally instead of reusing Health's Gene.
- There are no Source, Dataset, or Paper entities.
- There is no source provenance linking entities or relations back to BiGG, the model, or the paper.
- BiGG ID is useful, but it is overloaded across reactions, metabolites, genes, and compartments.
- Relation entities are mostly not named or provenance-backed in the way Disease Atlas evidence claims are.

Ontology inventory

Disease Atlas

Area	Terms in use
Shared Health/Root types	Disease, Symptom, Anatomical structure, Side effect, Drug, Drug class, Gene, Pathway, Dataset, Source, Paper
Local types	Biological Process, Molecular Function, Cellular Component
Core relation properties	Associated with, Treats, Related drugs, Related pathways, Sources, Includes, Presents with, Localizes to, Resembles, Palliates, Upregulates, Downregulates, plus GO annotation relations
Source and identifier fields	Website, DOI, Pmid, Publish date, Source database identifier, Version ID, DrugBank ID, Inchikey, GO ID, Disease Ontology ID, Entrez Gene ID, License

Ciaran's Space

Area	Terms in use
Local types	Reaction, Metabolite, Subsystem, Gene, Compartment
Relation properties	substrate of, product of, catalysed by, part of
Value properties	BiGG ID, Common Name, Formula, Charge, Gene Reaction Rule, Lower Bound, Upper Bound, Stoichiometry, Organism, Compartment Code
Source/provenance entities	None found
Pages/data blocks	None found

Live counts observed in Ciaran's space

The live API caps some result sets at 2,000, so counts marked as capped are lower bounds.

Type/property	Observed count	Notes
Reaction	At least 2,000	API-capped. Examples include exchange reactions and transport reactions.
Metabolite	1,844	Examples include palmitoyl-CoA, superoxide anion, arsenite, and other metabolites.
Gene	1,366	Local duplicate of Health Gene. Examples include mepM, lpxP, fhuF.
Subsystem	39	Examples include transport, membrane lipid metabolism, amino-acid metabolism, and unassigned.
Compartment	3	Cytoplasm, Periplasm, Extracellular.
Metabolic relation properties	At least 2,000 each for major relations	substrate of, product of, catalysed by, and part of are high-volume relation properties.

Direct overlap and reuse opportunities

There is almost no true ontology reuse overlap right now. The one obvious duplicate is:

Local Ciaran term	Existing reusable term	Recommendation
Gene	Health Gene = 93a0c3dc71314daf862e66c3af8b4fe4	Reuse Health Gene, or dual-type existing Ciaran gene entities with Health Gene and eventually retire the local type.

Other useful existing Root/Health terms Ciaran should reuse:

Need in Ciaran’s space	Existing Root/Health term
Dataset entity for model/source import	Dataset
Source database entity	Source
BiGG paper	Paper
Link every entity/relation to provenance	Sources
External URL	Website
Paper/source DOI	DOI
PubMed ID	Pmid
Publication/model date	Publish date
Database/model identifier	Source database identifier
Model/source version	Version ID
Gene synonym/locus tag if needed	Health Alias

There are also likely useful Health properties for metabolites if the BiGG import has cross-references: SMILES and KEGG ID.

Disease Atlas currently has local InChI and License. Those are useful ideas, but ideally they should become shared Root/Health-style properties rather than Ciaran reusing Disease Atlas-local properties.

Ciaran-specific terms to keep

I would not force everything into Disease Atlas or Health. These terms are domain-specific and should probably stay local, or become part of a future metabolism ontology.

Term	Keep?	Reason
Reaction	Yes	No Root/Health equivalent found. Core metabolic-model type.
Metabolite	Yes	No Root/Health equivalent found. Should maybe become a shared Health type later.
Compartment	Yes, but refine	Model compartment is not exactly the same as GO cellular component.
Subsystem	Maybe dual-type as Pathway	Many subsystems are pathway-like, but some are model categories such as transport or unassigned.
substrate of / product of	Keep or invert	Very useful metabolic relation vocabulary. Could be improved for navigation.

Stoichiometry, Lower Bound, Upper Bound, Gene Reaction Rule | Keep | These are model/math-specific fields, not general health ontology.

Suggested improvements for Ciaran’s space

Pass 1: provenance/source repair

Create:

- Dataset: likely BiGG model: E. coli K-12 MG1655, with exact model ID once confirmed.
- Source: BiGG Models.
- Paper: BiGG Models: A platform for integrating, standardizing and sharing genome-scale models.
- DOI: 10.1093/nar/gkv1049.
- Pmid: 26476456.
- Website: https://bigg.ucsd.edu/.
- Model page link if confirmed, likely https://bigg.ucsd.edu/models/iJO1366.

Then add Sources relations from reactions, metabolites, genes, compartments, subsystems, and relation/equation claim entities back to the dataset/source.

Pass 2: ontology reuse

- Add Health Gene type to all current gene entities.
- Keep BiGG ID, but also add generic Root Source database identifier.
- Add Website links on entities where possible: /models/{model}/reactions/{bigg_id}, /models/{model}/metabolites/{bigg_id}, /models/{model}/genes/{bigg_id}.
- Add Version ID to the dataset if the model or BiGG version is known.
- Use Alias for alternate gene names/locus tags if available.

Pass 3: relation cleanup

- Rename or normalize catalysed by to Catalyzed by or Catalyzed by gene.
- Normalize part of to Part of, if keeping it generic.
- Consider changing relation direction for reaction navigation: instead of Metabolite substrate of Reaction, use Reaction consumes Metabolite; instead of Metabolite product of Reaction, use Reaction produces Metabolite.
- Preserve stoichiometry on relation entities, not just as anonymous values.
- Name relation entities like Dodecanoate exchange - consumes - Dodecanoate.

Best way to make both ontologies more overlapping

The common layer should be provenance and identifiers first, biology second.

Shared/common ontology both spaces should use: Gene, Dataset, Source, Paper, Sources, Website, DOI, Pmid, Publish date, Source database identifier, Version ID, Alias.

Then, over time, promote or share these across spaces: Metabolite, Reaction, Catalyzed by, Consumes, Produces, Part of, Stoichiometry, Organism / Taxon, BiGG ID or source-specific identifier properties.

Main recommendation

Do not make Disease Atlas absorb Ciaran’s metabolism ontology yet. Instead, make both spaces interoperable through shared Gene, shared provenance, shared source identifiers, and source-backed relation entities.

That gives a clean bridge: Disease -> Gene in Disease Atlas; Gene -> Reaction -> Metabolite in Ciaran’s space.

Once Ciaran’s genes reuse Health Gene, a future dashboard can click a gene from Disease Atlas and discover metabolic reactions from Ciaran’s graph without merging the two spaces or overloading Disease Atlas with metabolic-model details.

Source notes

- Geo Disease Atlas space: https://www.geobrowser.io/space/141d3ace705feabc04d50c78bbf7226e
- Ciaran’s Geo space: https://www.geobrowser.io/space/aec359295ff90d00aea6696f0ddb1bbc
- BiGG Models: https://bigg.ucsd.edu/models
- BiGG Models paper: PubMed PMID 26476456, DOI 10.1093/nar/gkv1049